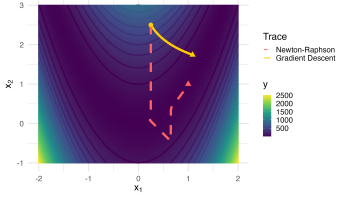
Optimization in Machine Learning

Second order methods
Newton-Raphson



Learning goals

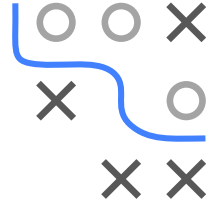
- Newton-Raphson
- Limitations

FROM FIRST TO SECOND ORDER METHODS

- So far: **First order methods**
  $\Rightarrow$ *Gradient* information, i.e., first derivatives

- Now: **Second order methods**
  $\Rightarrow$ *Hessian* information, i.e., second derivatives

NEWTON-RAPHSON



- **Assumption:** $f \in \mathcal{C}^2$
- **Aim:** Find stationary point $\boldsymbol{x}^*$, i.e., $\nabla f(\boldsymbol{x}^*) = \mathbf{0}$
- **Idea:** Find root of first order Taylor approx of $\nabla f(\boldsymbol{x})$:

$$\nabla f(\boldsymbol{x}) \approx \nabla f(\boldsymbol{x}^{[t]}) + \nabla^2 f(\boldsymbol{x}^{[t]})(\boldsymbol{x} - \boldsymbol{x}^{[t]}) = \mathbf{0}$$

$$\nabla^2 f(\boldsymbol{x}^{[t]})(\boldsymbol{x} - \boldsymbol{x}^{[t]}) = -\nabla f(\boldsymbol{x}^{[t]})$$

$$\boldsymbol{x}^{[t+1]} = \boldsymbol{x}^{[t]} - \left(\nabla^2 f(\boldsymbol{x}^{[t]})\right)^{-1} \nabla f(\boldsymbol{x}^{[t]})$$

- **Update scheme:**

$$\boldsymbol{x}^{[t+1]} = \boldsymbol{x}^{[t]} + \mathbf{d}^{[t]}$$

with $\mathbf{d}^{[t]} = -\left(\nabla^2 f(\boldsymbol{x}^{[t]})\right)^{-1} \nabla f(\boldsymbol{x}^{[t]})$

NEWTON-RAPHSON

- **Note:** In practice, we get $\mathbf{d}^{[t]}$ by solving the linear system

$$\nabla^2 f(\boldsymbol{x}^{[t]})\mathbf{d}^{[t]} = -\nabla f(\boldsymbol{x}^{[t]})$$

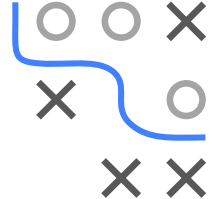
  with direct (matrix decompositions) or iterative methods

- **Relaxed/Damped Newton-Raphson:** Use step size $\alpha > 0$ with

$$\boldsymbol{x}^{[t+1]} = \boldsymbol{x}^{[t]} + \alpha \mathbf{d}^{[t]}$$

  to satisfy Wolfe conditions (or just Armijo rule)

ANALYTICAL EXAMPLE WITH QUADRATIC FORM



- Consider $f(x_1, x_2) = x_1^2 + \frac{x_2^2}{2}$
- Update direction: $\mathbf{d}^{[t]} = -\left(\nabla^2 f(x_1^{[t]}, x_2^{[t]})\right)^{-1} \nabla f(x_1^{[t]}, x_2^{[t]})$
- Gradient and Hessian:

$$\nabla f(x_1, x_2) = \begin{pmatrix} 2x_1 \\ x_2 \end{pmatrix}, \quad \nabla^2 f(x_1, x_2) = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$$

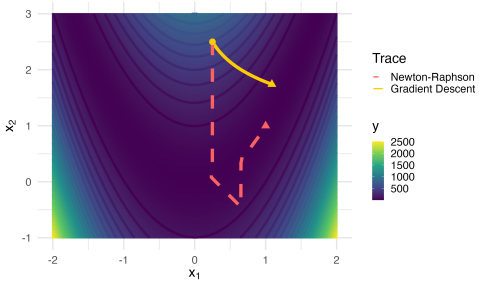
- First step:

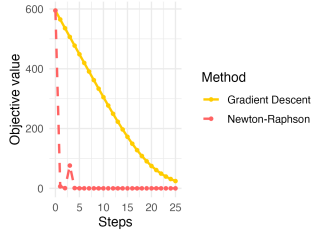
$$\begin{pmatrix} x_1^{[1]} \\ x_2^{[1]} \end{pmatrix} = \begin{pmatrix} x_1^{[0]} \\ x_2^{[0]} \end{pmatrix} + \mathbf{d}^{[0]} = \begin{pmatrix} x_1^{[0]} \\ x_2^{[0]} \end{pmatrix} - \begin{pmatrix} 1/2 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2x_1^{[0]} \\ x_2^{[0]} \end{pmatrix}$$

$$= \begin{pmatrix} x_1^{[0]} \\ x_2^{[0]} \end{pmatrix} + \begin{pmatrix} -x_1^{[0]} \\ -x_2^{[0]} \end{pmatrix} = \mathbf{0}$$

- **Note:** Newton-Raphson only needs one iteration for quadratic forms

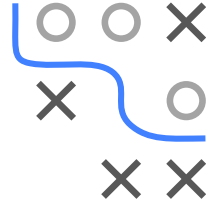
NEWTON-RAPHSON VS. GD ON BRANIN FUNCTION





Newton-Raphson has much better convergence speed here

DISCUSSION



Advantage:

- For f sufficiently smooth:

Newton-Raphson converges *locally* quadratically (i.e., for starting points close enough to stationary point)

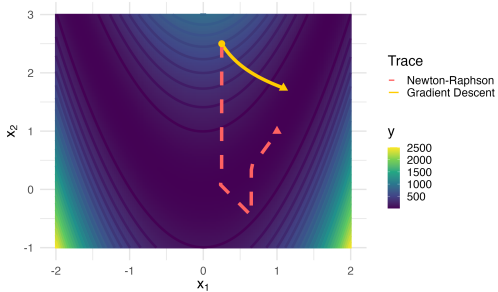
Disadvantage:

- For “bad” starting points:

Newton-Raphson may diverge

LIMITATIONS

- **Problem 1:** In general, $\mathbf{d}^{[t]}$ is not a descent direction

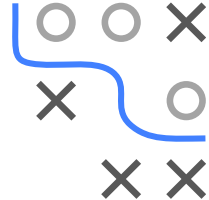


- **But**: If Hessian is positive definite, $\mathbf{d}^{[t]}$ is descent direction:

$$\nabla f(\boldsymbol{x}^{[t]})^\top \mathbf{d}^{[t]} = -\nabla f(\boldsymbol{x}^{[t]})^\top \left(\nabla^2 f(\boldsymbol{x}^{[t]})\right)^{-1} \nabla f(\boldsymbol{x}^{[t]}) < 0$$

- Near minimum, Hessian is positive definite
- For initial steps, Hessian is often not positive definite and Newton-Raphson may give non-descending update directions

LIMITATIONS



- **Problem 2:** Hessian can be **computationally expensive** to calculate, since descent direction $\mathbf{d}^{[t]}$ is the solution of the linear system

$$\nabla^2 f(\boldsymbol{x}^{[t]})\mathbf{d}^{[t]} = -\nabla f(\boldsymbol{x}^{[t]})$$

- **Aim:** Find quasi-second order methods not relying on exact Hessians
- Quasi-Newton method
- Gauss-Newton algorithm (for least squares)